

Лабораторная работа 14

Администрирование локальных сетей

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1 Введение

Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи

2 Выполнение лабораторной работы

Настройка интерфейсов коммутатора provider-sw-1 (рис. 2.1).

```

provider-archimede-sw-1

Physical Config CLI Attributes

IOS Command Line Interface

VLAN#3-CHANGED: Interface Vlan3, changed state to up
VLAN#5-CHANGED: Interface FastEthernet0/1, changed state to up
VLAN#PROT-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
VLAN#PROT-3-UPDOWN: Line protocol on Interface Vlan3, changed state to up
VLAN#5-CHANGED: Interface FastEthernet0/3, changed state to up
VLAN#PROT-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
VLAN#5-CHANGED: Interface FastEthernet0/4, changed state to up
VLAN#PROT-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
VLAN#3-UPDOWN: Interface FastEthernet0/4, changed state to down
VLAN#PROT-3-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
VLAN#5-CHANGED: Interface FastEthernet0/3, changed state to up
VLAN#PROT-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
VLAN#3-UPDOWN: Interface FastEthernet0/3, changed state to down
VLAN#PROT-3-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

Peer Address Verification
Password:
Authtypeable
Password:
Switch(configure terminal)
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface r0/3
Switch(config-if)#ip address mode trunk
Switch(config-if)#exit
Switch(config)#interface r0/4
Switch(config-if)#ip address mode trunk
Switch(config-if)#exit
Switch(config)#vlan 5
Switch(config-vlan)#name r12
Switch(config-vlan)#exit
Switch(config)#interface vlan3
Switch(config-if)#
VLAN#5-CHANGED: Interface Vlan3, changed state to up
VLAN#PROT-5-UPDOWN: Line protocol on Interface Vlan3, changed state to up
on shutdown
Switch(config-if)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name archide
Switch(config-vlan)#exit
Switch(config)#interface vlan4
Switch(config-if)#
VLAN#4-CHANGED: Interface Vlan4, changed state to up
}
VLAN#PROT-4-UPDOWN: Line protocol on Interface Vlan4, changed state to up

```

Рис. 2.1: Настройка provider-sw-1

Настройка интерфейсов маршрутизатора msk-donskaya-gw-1(рис. 2.2).

```
msk-donskaya-arromichina-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
since 2811 (MPC8540) processor (revision 0x200) with 6416K/5120K bytes of memory
Processor board ID JAD0519CMTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

User Access Verification

Password:

msk-donskaya-arromichina-gw-1>enable
Password:
msk-donskaya-arromichina-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arromichina-gw-1(config)#interface f0/1.5
msk-donskaya-arromichina-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
encapsulation dot1Q 5
msk-donskaya-arromichina-gw-1(config-subif)#ip address 10.128.255.1 255.255.255.252
msk-donskaya-arromichina-gw-1(config-subif)#description q42
msk-donskaya-arromichina-gw-1(config-subif)#exit
msk-donskaya-arromichina-gw-1(config)#interface f0/1.6
msk-donskaya-arromichina-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up
encapsulation dot1Q 6
msk-donskaya-arromichina-gw-1(config-subif)#ip address 10.128.255.5 255.255.255.252
msk-donskaya-arromichina-gw-1(config-subif)#description sock1
msk-donskaya-arromichina-gw-1(config-subif)#exit
msk-donskaya-arromichina-gw-1(config)#exit
%SYS-5-CONFIG I: Configured from console by console
write memory
Building configuration...
[OK]
msk-donskaya-arromichina-gw-1#
```

Рис. 2.2: Настройка msk-donskaya-gw-1

Настройка интерфейсов маршрутизатора msk-q42-gw-1 (рис. 2.3).

```
msk-q42-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wvl/export/crypto/tool/atqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD0519GMTZ (4292891495)
4 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

User Access Verification

Password:

msk-q42-gw-1>enable
Password:
msk-q42-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-gw-1(config)#interface f0/1.5
msk-q42-gw-1(config-subif)#encapsulation dot1Q 5
msk-q42-gw-1(config-subif)#exit
msk-q42-gw-1(config)#interface f0/1
msk-q42-gw-1(config-if)#no shutdown

msk-q42-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
exit
msk-q42-gw-1(config)#interface f0/1.5
msk-q42-gw-1(config-subif)#encapsulation dot1Q 5
msk-q42-gw-1(config-subif)#ip address 10.128.255.2 255.255.255.252
msk-q42-gw-1(config-subif)#description donakaya
msk-q42-gw-1(config-subif)#exit
msk-q42-gw-1(config)#exit
msk-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-q42-gw-1#
```

Рис. 2.3: Настройка msk-q42-gw-1

Настройка интерфейсов коммутатора sch-sochi-sw-1 (рис. 2.4).


```
sch-sochi-sw-1
Physical Config CLI Attributes
IOS Command Line Interface

Cisco WS-C2950-24 (RC32300) processor (revision C0) with 21039K bytes of memory.
Processor board ID FHK0610Z0MC
Running Standard Image
24 FastEthernet/IEEE 802.3 interface(s)
63488K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address: 00D0.5786.3980
Motherboard assembly number: 73-5781-09
Power supply part number: 34-0945-01
Motherboard serial number: FOC0610045Z
Power supply serial number: DAB0609127D
Model revision number: C0
Motherboard revision number: A0
Model number: WS-C2950-24
System serial number: FHK0610Z0MC

Cisco Internetwork Operating System Software
IOS (tm) C2950 Software (C2950-I6Q4L2-M), Version 12.1(22)EA4, RELEASE SOFTWARE(fx)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 18-May-05 22:31 by jharirba

Press RETURN to get started!

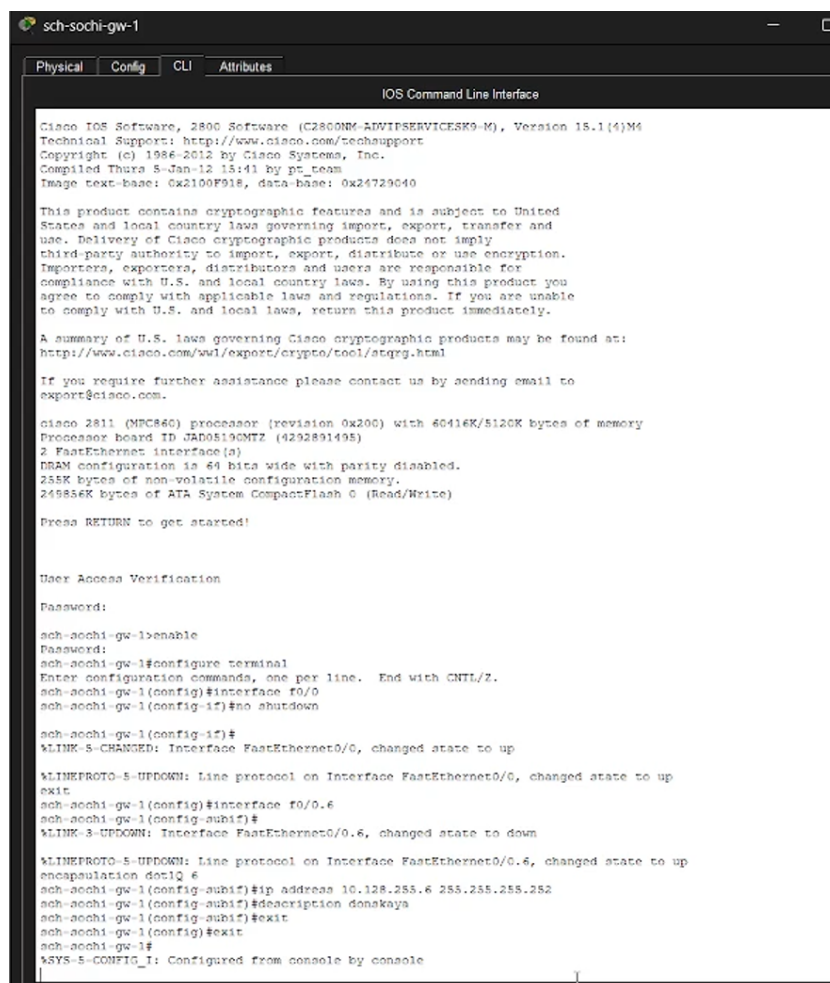
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

User Access Verification
Password:
sch-acchi-aw-1benable
Password:
sch-acchi-aw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-acchi-aw-1(config)#interface f0/23
sch-acchi-aw-1(config-if)#switchport mode trunk
sch-acchi-aw-1(config-if)#exit
sch-acchi-aw-1(config)#interface f0/24
sch-acchi-aw-1(config-if)#switchport mode trunk

sch-acchi-aw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
exit
sch-acchi-aw-1(config)#vlan 6
sch-acchi-aw-1(config-vlan)#name sochi
sch-acchi-aw-1(config-vlan)#exit
sch-acchi-aw-1(config)#interface vlan6
sch-acchi-aw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
no shutdo
```

Рис. 2.4: Настройка sch-sochi-sw-1

Настройка интерфейсов маршрутизатора sch-sochi-gw-1 (рис. 2.5).



```
sch-sochi-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 15.1(4)M4
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thu Jan 12 15:41 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729C40

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wd/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interfaces (a)
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

User Access Verification

Password:

sch-sochi-gw-1>enable
Password:
sch-sochi-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gw-1(config)#interface f0/0
sch-sochi-gw-1(config-if)#no shutdown

sch-sochi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
sch-sochi-gw-1(config)#interface f0/0.6
sch-sochi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up
encapsulation dot1Q 6
sch-sochi-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
sch-sochi-gw-1(config-subif)#description donakaya
sch-sochi-gw-1(config-subif)#exit
sch-sochi-gw-1(config)#exit
sch-sochi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Рис. 2.5: Настройка sch-sochi-gw-1

Настройка интерфейсов маршрутизатора msk-q42-gw-1 (рис. 2.6).

```
msk-q42-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
LINK-3-CHANGED: Interface FastEthernet0/1, changed state to up
MLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down
MLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
msk-q42-gw-1(config)#interface f0/1.5
msk-q42-gw-1(config-subif)#encapsulation dot1q 5
msk-q42-gw-1(config-subif)#ip address 10.128.255.2 255.255.255.252
msk-q42-gw-1(config-subif)#description dmskaya
msk-q42-gw-1(config-subif)#exit
msk-q42-gw-1(config)#exit
msk-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-q42-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-gw-1(config)#interface f0/0
msk-q42-gw-1(config-if)#no shutdown

msk-q42-gw-1(config-if)#
LINK-3-CHANGED: Interface FastEthernet0/0, changed state to up
MLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
msk-q42-gw-1(config)#interface f0/0.201
msk-q42-gw-1(config-subif)#
LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down
MLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up
encapsulation dot1q 201
msk-q42-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
msk-q42-gw-1(config-subif)#description q42-main
msk-q42-gw-1(config-subif)#exit
msk-q42-gw-1(config)#interface f1/0
msk-q42-gw-1(config-if)#no shutdown

msk-q42-gw-1(config-if)#
LINK-3-CHANGED: Interface FastEthernet1/0, changed state to up
MLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
msk-q42-gw-1(config)#interface f1/0.202
msk-q42-gw-1(config-subif)#
LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down
MLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up
encapsulation dot1q 202
msk-q42-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
msk-q42-gw-1(config-subif)#description q42-management
msk-q42-gw-1(config-subif)#exit
msk-q42-gw-1(config)#exit
msk-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-q42-gw-1#
```

Рис. 2.6: Настройка msk-q42-gw-1

Настройка интерфейсов коммутатора msk-q42-sw-1 (рис. 2.7).

```
msk-q42-sw-1
Physical Config CLI Attributes
IOS Command Line Interface

Running Standard Image
24 FastEthernet/IEEE 802.3 interface(s)
63488K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address: 0001.8643.8E7B
Motherboard assembly number: 73-5781-09
Power supply part number: 34-0945-01
Motherboard serial number: FOC0610045Z
Power supply serial number: DAB0609127D
Model revision number: C0
Motherboard revision number: A0
Model number: KS-C2950-24
System serial number: THK061020WC

Cisco Internetwork Operating System Software
IOS (tm) C2950 Software (C2950-I6Q4L2-M), Version 12.1(22)EA4, RELEASE SOFTWARE(fol)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 18-May-05 22:13 by jharirba

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

User Access Verification
Password:
msk-q42-sw-1>enable
Password:
msk-q42-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-sw-1(config)#interface f0/24
msk-q42-sw-1(config-if)#switchport mode trunk

msk-q42-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
exit
msk-q42-sw-1(config)#interface f0/1
msk-q42-sw-1(config-if)#switchport mode access
msk-q42-sw-1(config-if)#switchport access vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-sw-1(config-if)#exit
msk-q42-sw-1(config)#vlan 201
msk-q42-sw-1(config-vlan)#name q42-main
msk-q42-sw-1(config-vlan)#exit
msk-q42-sw-1(config)#interface vlan201
msk-q42-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up
no shutdown
msk-q42-sw-1(config-if)#ex
```

Рис. 2.7: Настройка msk-q42-sw-1

Настройка интерфейсов маршрутизирующего коммутатора msk-hostel-gw-1 (рис. 2.8).

```
msk-hostel-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

User Access Verification

Password:

msk-hostel-gw-1enable
Password:
msk-hostel-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-gw-1(config)#interface g0/1
msk-hostel-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-gw-1(config-if)#switchport mode trunk
...
msk-hostel-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
msk-hostel-gw-1(config)#interface f0/1
msk-hostel-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-gw-1(config-if)#switchport mode trunk
...
msk-hostel-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
msk-hostel-gw-1(config)#vlan 202
msk-hostel-gw-1(config-vlan)#name q42-management
msk-hostel-gw-1(config-vlan)#exit
msk-hostel-gw-1(config)#interface vlan202
msk-hostel-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up
no shutdown
msk-hostel-gw-1(config-if)#ip address 10.129.1.2 255.255.255.0
msk-hostel-gw-1(config-if)#exit
msk-hostel-gw-1(config)#vlan 301
msk-hostel-gw-1(config-vlan)#name hostel-main
msk-hostel-gw-1(config-vlan)#exit
msk-hostel-gw-1(config)#interface vlan301
msk-hostel-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
no shutdown
msk-hostel-gw-1(config-if)#ip address 10.129.128.1 255.255.255.0
msk-hostel-gw-1(config-if)#exit
msk-hostel-gw-1(config)#exit
msk-hostel-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-hostel-gw-1#
```

Рис. 2.8: Настройка msk-hostel-gw-1

Настройка интерфейсов коммутатора msk-hostel-sw-1 (рис.2.9).

```
msk-hostel-sw-1
Physical Config CLI Attributes
IOS Command Line Interface
Cisco WS-C2950-24 (M332300) processor (revision 10) with 21038K bytes of memory.
Processor board ID FTK0610Z0WC
Running Standard Image
24 FastEthernet/IEEE 802.3 interface(s)
2 Gigabit Ethernet/IEEE 802.3 interface(s)

63488K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address: 0040.0890.0DEC
Motherboard assembly number: 73-5781-05
Power supply part number: 34-0945-01
Motherboard serial number: FOC0610045Z
Power supply serial number: DAR0609127D
Model revision number: C0
Motherboard revision number: A0
Model number: WS-C2950T-24
System serial number: FTK0610Z0WC

Cisco Internetwork Operating System Software
IOS (tm) C2950 Software (C2950-I6Q4L2-M), Version 12.1(22)EA4, RELEASE SOFTWARE(fx)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 18-May-05 22:31 by jharirba

Press RETURN to get started!

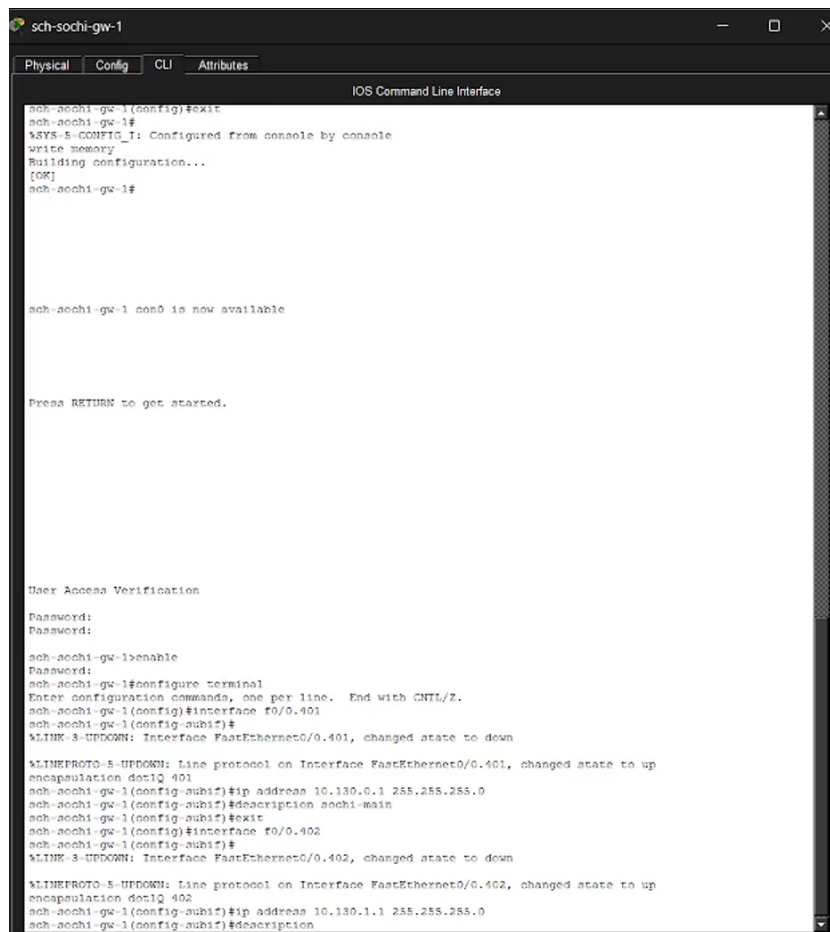
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk GigabitEthernet0/1 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

User Access Verification
Password:
msk-hostel-sw-1enable
Password:
msk-hostel-sw-1configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-sw-1(config)#interface g0/1
msk-hostel-sw-1(config-if)#switchport mode trunk
msk-hostel-sw-1(config-if)#exit
msk-hostel-sw-1(config)#interface f0/1
msk-hostel-sw-1(config-if)#switchport mode access
msk-hostel-sw-1(config-if)#switchport access vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-sw-1(config-if)#exit
msk-hostel-sw-1(config)#vlan 301
msk-hostel-sw-1(config-vlan)#name hostel-main
msk-hostel-sw-1(config-vlan)#exit
msk-hostel-sw-1(config)#interfa
```

Рис. 2.9: Настройка msk-hostel-sw-1

Настройка интерфейсов маршрутизатора sch-sochi-gw-1 (рис. 2.10).



```
sch-sochi-gw-1 (config)#exit
sch-sochi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
sch-sochi-gw-1#

sch-sochi-gw-1 con0 is now available

Press RETURN to get started.

User Access Verification

Password:
Password:

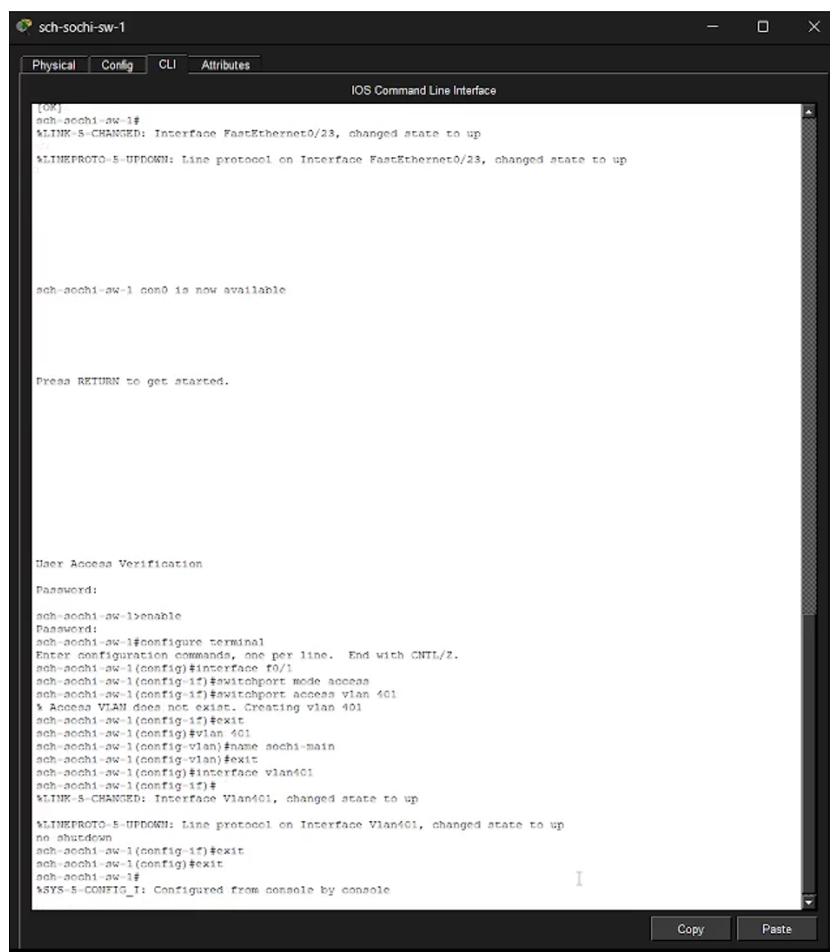
sch-sochi-gw-1 enable
Password:
Password:
sch-sochi-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gw-1 (config)#interface f0/0.401
sch-sochi-gw-1 (config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINKPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up
encapsulation dot1Q 401
sch-sochi-gw-1 (config-subif)#ip address 10.130.0.1 255.255.255.0
sch-sochi-gw-1 (config-subif)#description sochi-main
sch-sochi-gw-1 (config-subif)#exit
sch-sochi-gw-1 (config)#interface f0/0.402
sch-sochi-gw-1 (config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

%LINKPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up
encapsulation dot1Q 402
sch-sochi-gw-1 (config-subif)#ip address 10.130.1.1 255.255.255.0
sch-sochi-gw-1 (config-subif)#description
```

Рис. 2.10: Настройка sch-sochi-gw-1

Настройка интерфейсов коммутатора sch-sochi-sw-1 (рис. 2.11).



```
[OK]
sch-acchi-sw-1#
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

sch-acchi-sw-1 con0 is now available

Press RETURN to get started.

User Access Verification
Password:
sch-acchi-sw-1>enable
Password:
sch-acchi-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-acchi-sw-1(config)#interface f0/1
sch-acchi-sw-1(config-if)#switchport mode access
sch-acchi-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-acchi-sw-1(config-if)#exit
sch-acchi-sw-1(config)#vlan 401
sch-acchi-sw-1(config-vlan)#name acchi-main
sch-acchi-sw-1(config-vlan)#exit
sch-acchi-sw-1(config)#interface Vlan401
sch-acchi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up
sch-acchi-sw-1(config-if)#exit
sch-acchi-sw-1(config)#exit
sch-acchi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Рис. 2.11: Настройка sch-sochi-sw-1

Настройка маршрутизатора msk-donskaya-gw-1 (рис. 2.12).

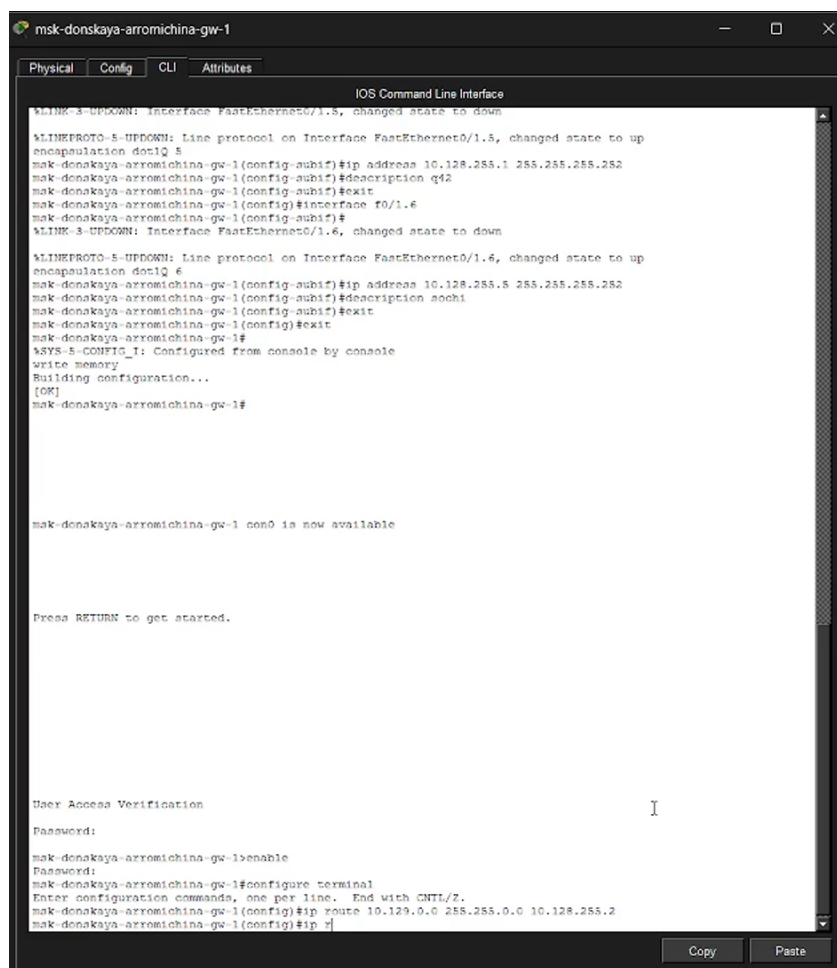


Рис. 2.12: Настройка msk-donskaya-gw-1

Настройка маршрутизатора msk-q42-gw-1 (рис. 2.13).

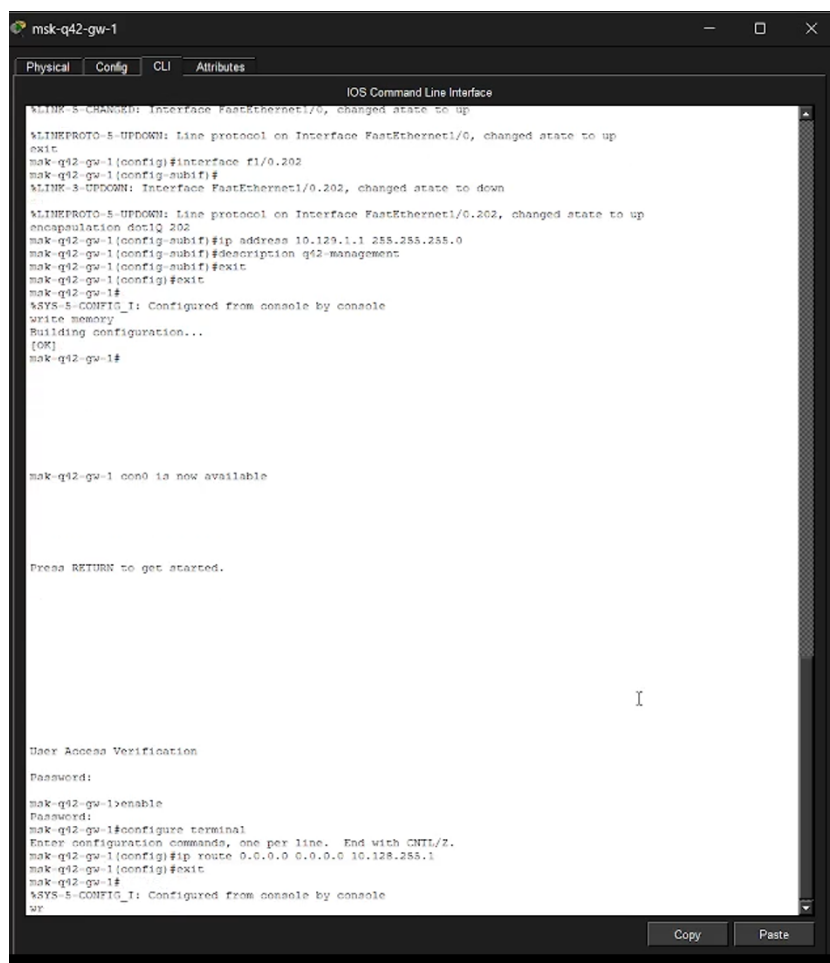


Рис. 2.13: Настройка msk-q42-gw-1

Настройка маршрутизатора sch-sochi-gw-1 (рис. 2.14).

```
sch-sochi-gw-1 con0 is now available

Press RETURN to get started.

User Access Verification
Password:
Password:

sch-sochi-gw-1>enable
Password:
sch-sochi-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gw-1(config)#interface f0/0.401
sch-sochi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up
sch-sochi-gw-1(config-subif)#ip address 10.130.0.1 255.255.255.0
sch-sochi-gw-1(config-subif)#description sochi-main
sch-sochi-gw-1(config-subif)#exit
sch-sochi-gw-1(config)#interface f0/0.402
sch-sochi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up
sch-sochi-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
sch-sochi-gw-1(config-subif)#description sochi-management
sch-sochi-gw-1(config-subif)#exit
sch-sochi-gw-1(config)#exit
sch-sochi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
sch-sochi-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-gw-1(config)#
```

Рис. 2.14: Настройка sch-sochi-gw-1

Настройка маршрутизатора msk-q42-gw-1 (рис. 2.15).

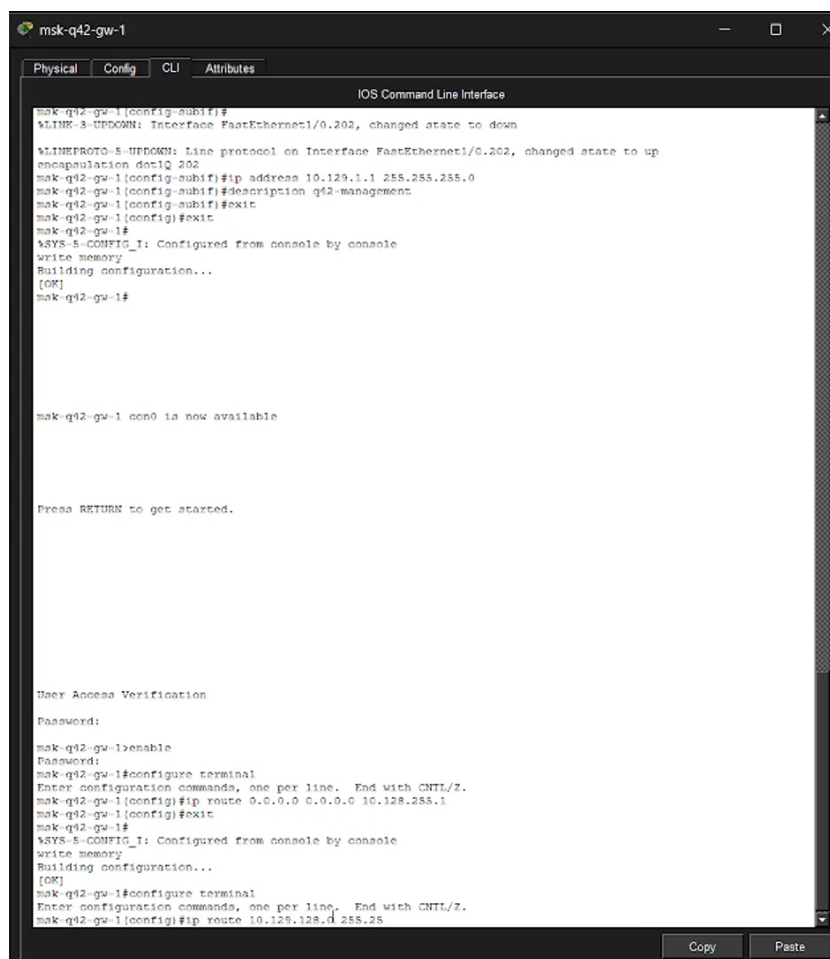


Рис. 2.15: Настройка msk-q42-gw-1

Настройка интерфейсов маршрутизирующего коммутатора msk-hostel-gw-1 (рис. 2.16).

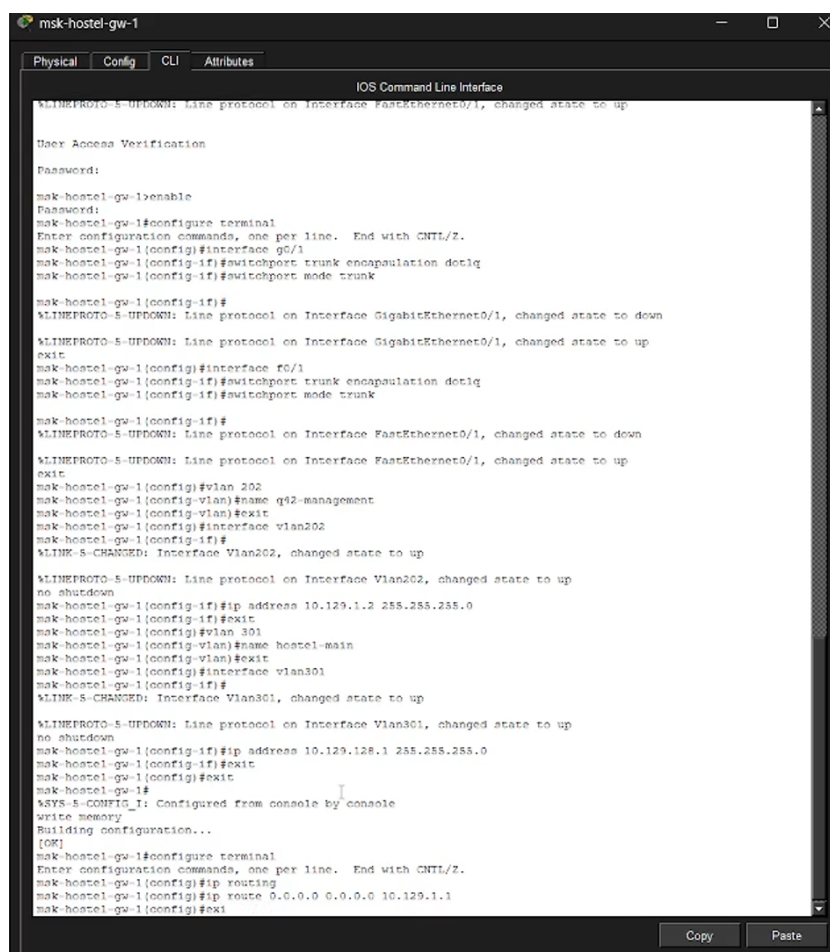


Рис. 2.16: Настройка msk-hostel-gw-1

Настройка NAT на маршрутизаторе msk-donskaya-gw-1 (рис. 2.17).

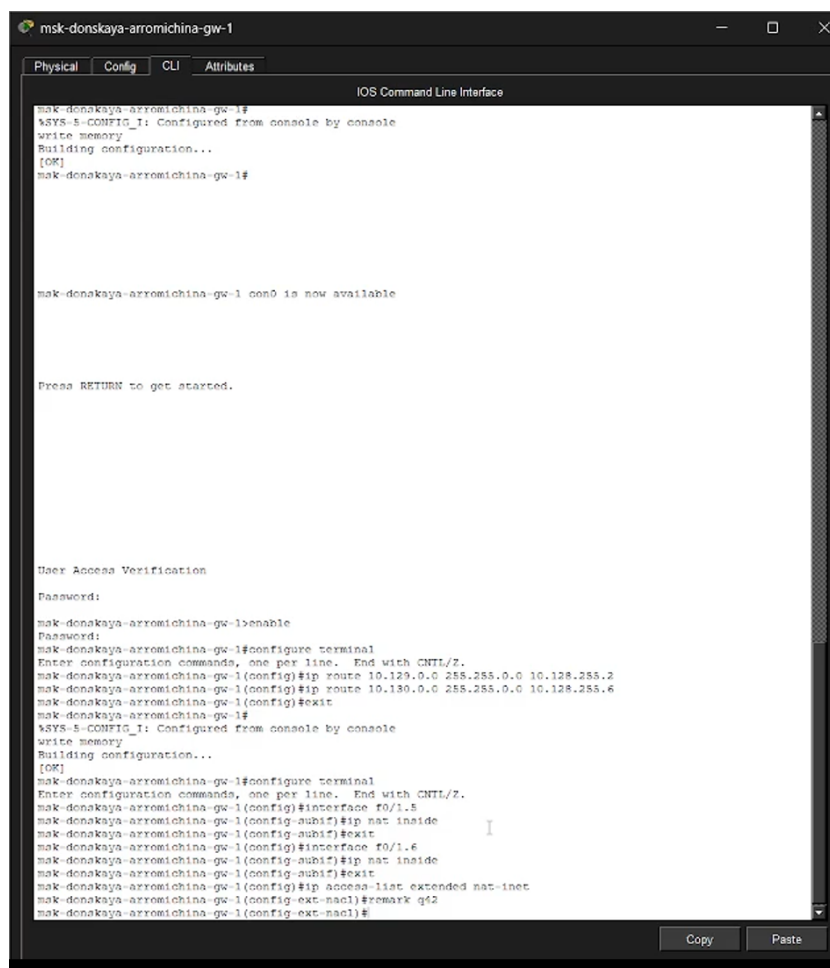


Рис. 2.17: Настройка msk-donskaya-gw-1

3 Вывод

Мы смогли настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи

4 Контрольные вопросы

1. Пример из файла (раздел 14.3.4.3) — настройка маршрута по умолчанию на маршрутизаторе sch-sochi-gw-1:

```
text sch-sochi-gw-1>enable sch-sochi-gw-1#configure terminal sch-sochi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
```

 Пояснение:

0.0.0.0 0.0.0.0 — сеть назначения и маска (маршрут по умолчанию).

10.128.255.5 — IP-адрес следующего шлюза (в данном случае msk-donskaya-gw-1).

Другой пример (гипотетический, между двумя подсетями организации, например, между VLAN 201 и VLAN 202): На маршрутизаторе msk-q42-gw-1:

```
text msk-q42-gw-1(config)#ip route 10.129.1.0 255.255.255.0 10.129.0.2
```

 Это означает, что для достижения подсети 10.129.1.0/24 трафик направляется на шлюз 10.129.0.2.

2. Опишите процесс обращения устройства из одного VLAN к устройству из другого VLAN. Отправка запроса: Устройство в VLAN A (IP: 10.129.0.10) пытается связаться с устройством в VLAN B (IP: 10.129.1.20). Поскольку IP-адреса находятся в разных подсетях, отправитель понимает, что цель находится не в его локальной сети.

ARP-запрос на шлюз по умолчанию: Отправитель отправляет ARP-запрос для MAC-адреса своего шлюза по умолчанию (например, 10.129.0.1 — это msk-q42-gw-1).

Передача пакета на маршрутизатор: Пакет (с исходным MAC отправителя и MAC шлюза) отправляется на маршрутизатор.

Маршрутизация между VLAN: Маршрутизатор (или многоуровневый коммутатор) получает пакет на субинтерфейсе .201 (VLAN 201), смотрит в таблицу маршрутизации и определяет, что сеть VLAN 202 доступна через субинтерфейс .202 (или другой интерфейс).

ARP со стороны маршрутизатора: Маршрутизатор выполняет ARP-запрос в VLAN 202 для определения MAC-адреса целевого устройства (10.129.1.20).

Пересылка пакета: Маршрутизатор заменяет исходный MAC на свой MAC в VLAN 202, MAC назначения — на MAC целевого устройства, и отправляет пакет в VLAN 202.

Ответ: Устройство в VLAN B получает пакет и отвечает аналогичным образом через маршрутизатор.

Важно: Без маршрутизатора (L3-устройства) устройства из разных VLAN не могут общаться напрямую, даже если они физически подключены к одному коммутатору.

3. Как проверить работоспособность маршрута? Основные способы:

Команда ping — проверка связности с удалённым устройством:

text ping 10.130.0.1 Успешные ответы (!!!!!) означают, что маршрут работает.

Команда traceroute (или tracert) — показывает путь следования пакетов:

text traceroute 10.130.0.1 Позволяет увидеть, через какие шлюзы проходит трафик, и где возможен обрыв.

Extended ping (в Cisco) с указанием источника:

text ping 10.130.0.1 source 10.129.0.1 Проверяет маршрут от конкретного интерфейса.

Проверка таблицы маршрутизации — убедиться, что маршрут внесён и имеет правильный next-hop (см. следующий вопрос).

4. Как посмотреть таблицу маршрутизации? На оборудовании Cisco (маршрутизаторы, многоуровневые коммутаторы):

Основная команда:

text show ip route Пример вывода (гипотетический):

text Codes: C - connected, S - static, O - OSPF, ... S 10.130.0.0/24 [1/0] via 10.128.255.6 C 10.128.255.4/30 is directly connected, FastEthernet0/1.6 S* 0.0.0.0/0 [1/0] via 10.128.255.5 S — статический маршрут

C — подключённая сеть

S* — маршрут по умолчанию

Другие полезные команды:

show ip route static — только статические маршруты

show ip route connected — только подключённые сети

show ip route 10.130.0.0 — детали конкретного маршрута

На компьютере (Windows):

text route print или

text netstat -r На Linux:

text ip route show или

text route -n

Список литературы